Nishkarsh Jain

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Objective

Aspiring VLSI engineer and a fresh graduate seeking a fresher opportunity. Through internships at Synopsys, Nvidia and Jabil, I have hands-on experience in Emulation, UART protocol, STA, Physical Design, Scripting and automation. I am committed to contributing to innovative projects by utilising my core knowledge and excellent problem-solving skills.

Education

THAPAR INSTITUTE OF ENGINEERING AND TECHNOLOGY — Patiala, Punjab
B.E., Electronics and Communication Engineering. CGPA-9.05 — 2021 - 2025
(Merit Scholarship - III awardee)

SILVER BELLS PUBLIC SCHOOL — Shamli, Uttar Pradesh
Senior Secondary Education(XII). Percentage-93.5% — 2018 - 2020

ST. FRANCIS SCHOOL — Shamli, Uttar Pradesh
Secondary Education(X). Percentage-93.8% — 2018

Experience

Synopsys India Pvt. Ltd. — Noida, Uttar Pradesh
Technical Engineer Apprentice — September 2025 - **Ongoing**

- Development of UART Transactor for Zebu emulator using C++
- Protocol functionality enhancements for Zebu Transactors using C++
- Development of HW/SW Testbench Backwards Compatibility Agent and Script for Zebu Transactors with added AI Code review

Nvidia Graphics Pvt. Ltd. — Bangalore, Karnataka
ASIC Engineer Intern (here) — January 2025 - July 2025

- Timing Closure and Performance Optimisation for Advanced SoC Designs using Static Timing Analysis (STA)
- Perl Scripts for Automating STA report analysis
- Manual fixes for clock transition and crosstalk violations

Jabil Circuit India Private Limited — Pune, Maharashtra
Test Engineer Intern (here) — June 2024 - July 2024

- Working, Debugging and Testing of Ericsson's 5G Radio Transceiver - AIR3268
- SMT Process of Ericsson AIR 3268 Transceiver PCBA
- Setup, measurement, and calibration of Keysight test equipment

Academic Projects

Development and Implementation of a 5G Modem using an FPGA Board — January 2024 - January 2025

- FM Broadcast and Radio using Adalm-Pluto SDR🔗
- Hardware-in-the-Loop Simulation of FFT on Kintex-7 KC705 FPGA Development Board
- Simulink Model Development, FPGA implementation of 5G transceiver using Vitis Model Composer

Robosoccer Bot🔗 — February 2024 - April 2024

- Built an RC-controlled bot using a laser-cut metal chassis, Sabertooth motor driver, and geared motors

Test Data Compression for System-on-a-Chip Using Huffman Codes — April 2024

- Implementation of Huffman Codes for data compression using Xilinx Vivado

Smart Irrigation System🔗 — June 2023 - July 2023

- Worked on a smart irrigation system for flood-irrigated paddy fields using IoT-based systems as group leader.
- Used ESP-32 microcontroller, nRF24L01 RF module, GSM SIM800A, and BME280 & Thingspeak IoT Cloud.
- Worked on water conservation by automating water flow and alerting the farmer about the weather.

Skills & Interests

Technical: Verilog, System Verilog, C/C++, PERL, Python, PCB design
Embedded Hardware: Adalm-Pluto SDR, KC705 FPGA board, STM32, ESP-32, Arduino, Sensors
Laboratory: Synopsys VCS, Synopsys Prime Time, Xilinx Vivado, Matlab, Simulink, S-Edit, Silvaco, Arduino IDE

Certifications

ZEBU Emulator Fundamental Training — 2025
VLSI Design-FPGA and ASIC (here**)** — 2023
IoT-Based Systems (here**)** — 2022 – 2023